Advancing the use of genomic assessments of local adaptation in applied conservation practice

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This afternoon's topics:

(Potential) applications of adaptive variation in applied conservation.

Is adaptive variation being incorporated into applied conservation?

Why aren't studies of adaptive variation (usually) being incorporated into applied conservation?

A few examples & why they're working

How do we continue to advance applications?

Red wolf (*Canis rufus*) - Endangered; photo: Dave Pape

(Potential) applications of adaptive variation in applied conservation

- Identify locally adapted populations
- Identify sources and targets for genetic rescue
- Identify populations at risk of future maladaptation and sources for assisted gene flow
- Identify large-effect loci contributing to conservation-relevant traits
- Monitor fitness-related alleles to evaluate impacts of conservation management actions

Is adaptive variation being incorporated into applied conservation?

Biological Conservation 299 (2024) 110788

Contents lists available at ScienceDirect

Biological Conservation

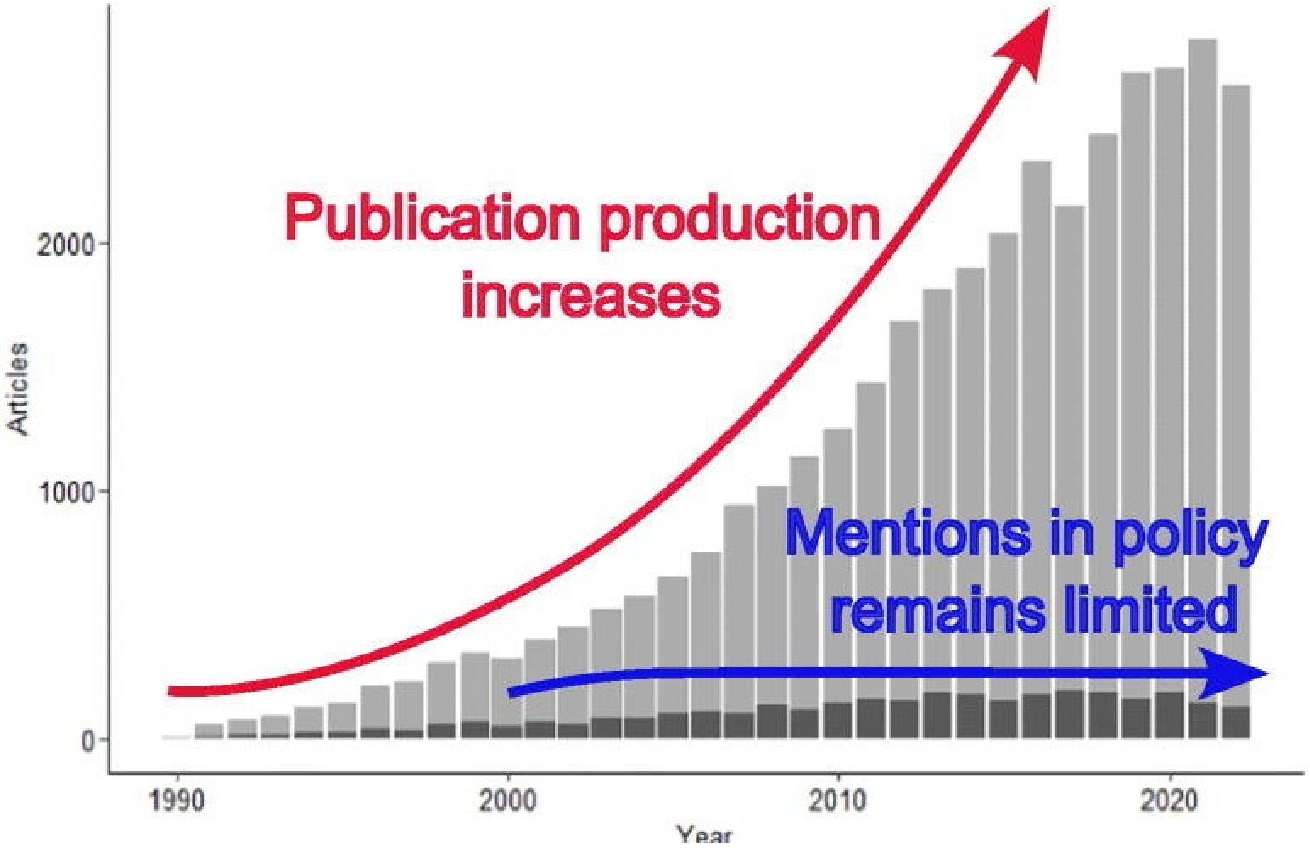
journal homepage: www.elsevier.com/locate/biocon

Technological advances have enhanced and expanded conservation genomics research but are yet to be integrated fully into biodiversity management

Linda E. Neaves [a,*], Brittany M. Brockett [a], Maldwyn J. Evans [a], Jennifer C. Pierson [a,b,c], Stephen D. Sarre [c]

Short answer: *not really…*

Is adaptive variation being incorporated into applied conservation?



>36,000 publications related to conservation genetics

1,548 related to "adaptation" (~4%)

- 92% had academic citations
- 12% had "practical" citations

Neaves et al. 2024, *Biological Conservation*

Is adaptive variation being incorporated into applied conservation?

MOLECULAR ECOLOGY

MINI REVIEW | Open Access

The Genomics Revolution in Nonmodel Species: Predictions vs. Reality for Salmonids

Samuel A. May, Samuel W. Rosenbaum, Devon E. Pearse, Marty Kardos, Craig R. Primmer, Diana S. Baetscher, Robin S. Waples

First published: 18 April 2025 | https://doi.org/10.1111/mec.17758

“The promise of genomics to change everything”

Is adaptive variation being incorporated into applied conservation?

Predicted & Realized: detection of adaptive diversity

- Applications tend to be in aquaculture (e.g., disease resistance)
- Informing assisted gene flow/migration has been limited
- Genomic offsets are promising but remain unvalidated & utility is uncertain *(also Layton et al. 2024, Global Change Biology)*
- Gene-targeted conservation is cautioned against (overlooks broader ecological and evolutionary processes)

May et al. 2025, *Molecular Ecology*

Is adaptive variation being incorporated into applied conservation?

Largely Unforeseen but Realized: prevalence of large-effect loci

- Adult migration timing – relevant for management but not directly incorporated (yet)

“Although there are few documented instances where genomic data have directly hindered conservation actions, there are cases where [these data have] introduced complexity to conservation conversations, delaying decision-making processes.”

May et al. 2025, *Molecular Ecology*

Is adaptive variation being incorporated into applied conservation?

Predicted & Unmet: delineation of conservation or evolutionary significant units

- New insights (relative to traditional approaches) have been limited but not completely absent
- Some conservation applications for intra and interspecific unit delineations
- Has sometimes created new challenges

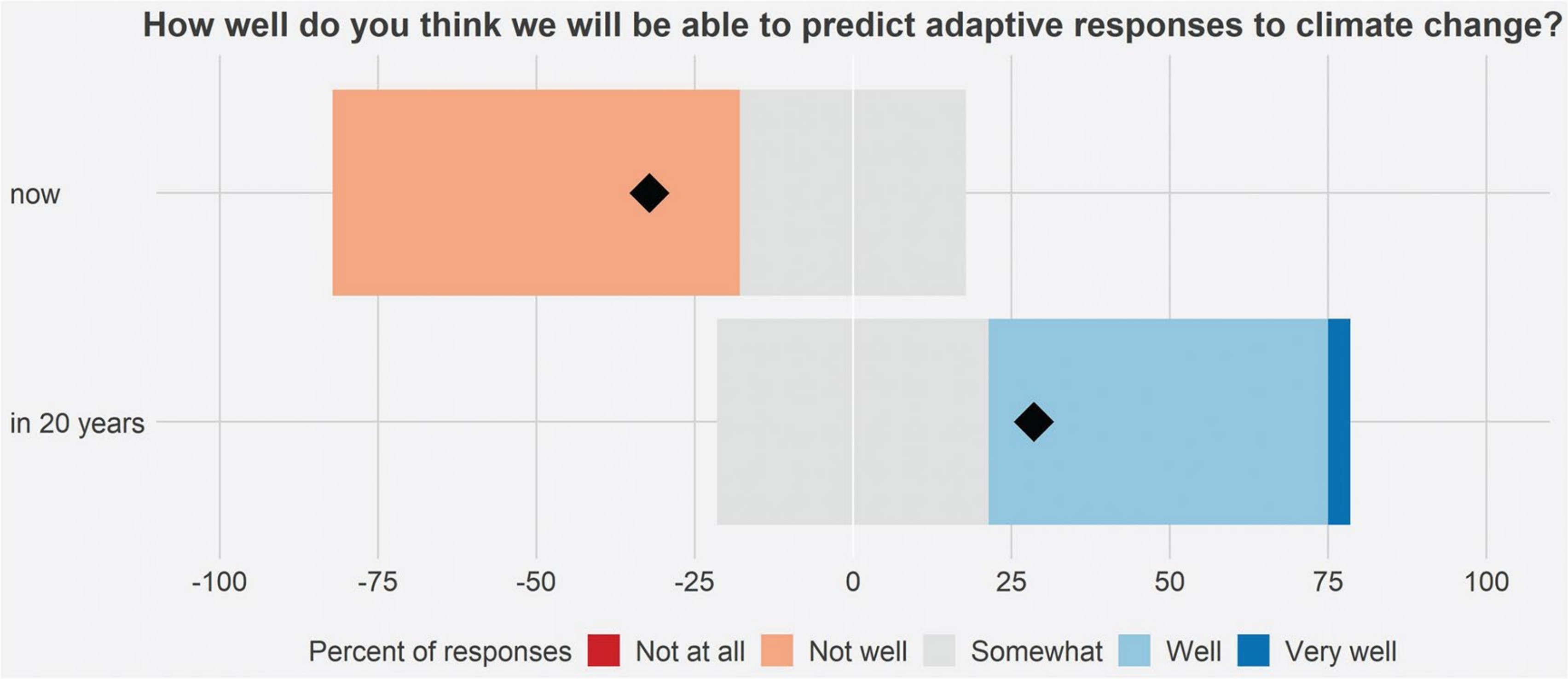
May et al. 2025, *Molecular Ecology*

Why aren't studies of adaptive variation being incorporated into applied conservation?

- Poor alignment between analyses and practitioner needs/priorities (i.e., results are not considered useful by practitioners and/or geneticists fail to provide analyses that are needed)
- Practitioners don't understand the data/results so ignore them (i.e., poor communication)
- Lack of coordination among academics, NGOs, state/provincial, and federal entities

Why aren't studies of adaptive variation being incorporated into applied conservation?

- Too much uncertainty (academics and practitioners)



Urban et al. 2024, *Evolution Letters*

(Potential) applications of adaptive variation in applied conservation

- Identify locally adapted populations
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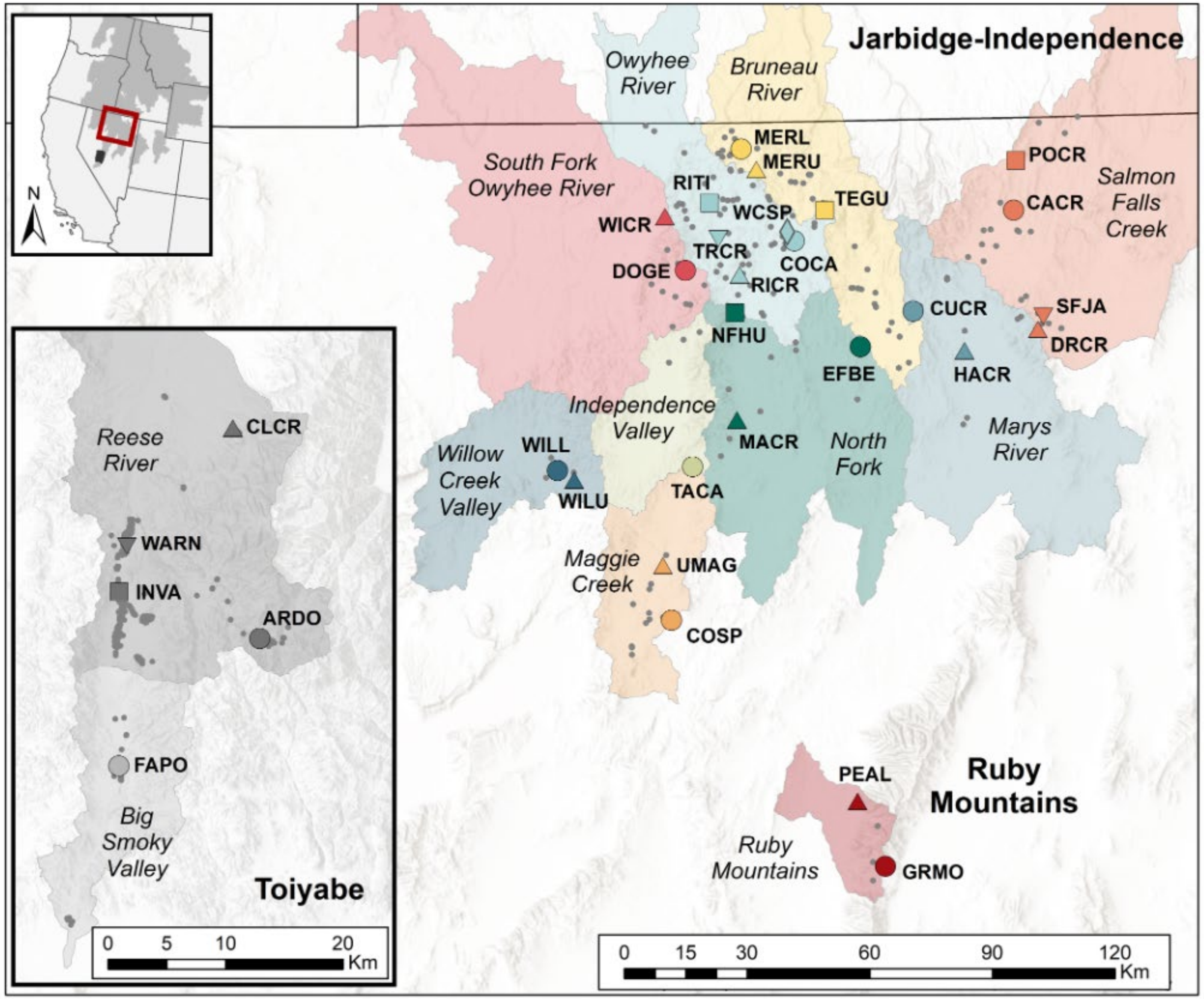
Examples: local adaptation and conservation units

Evolutionary significant units?

✓ Reproductive isolation

✗ Adaptive divergence

Photo: Joel Sartore



Forester et al. 2022, Molecular Ecology

Examples: genetic rescue

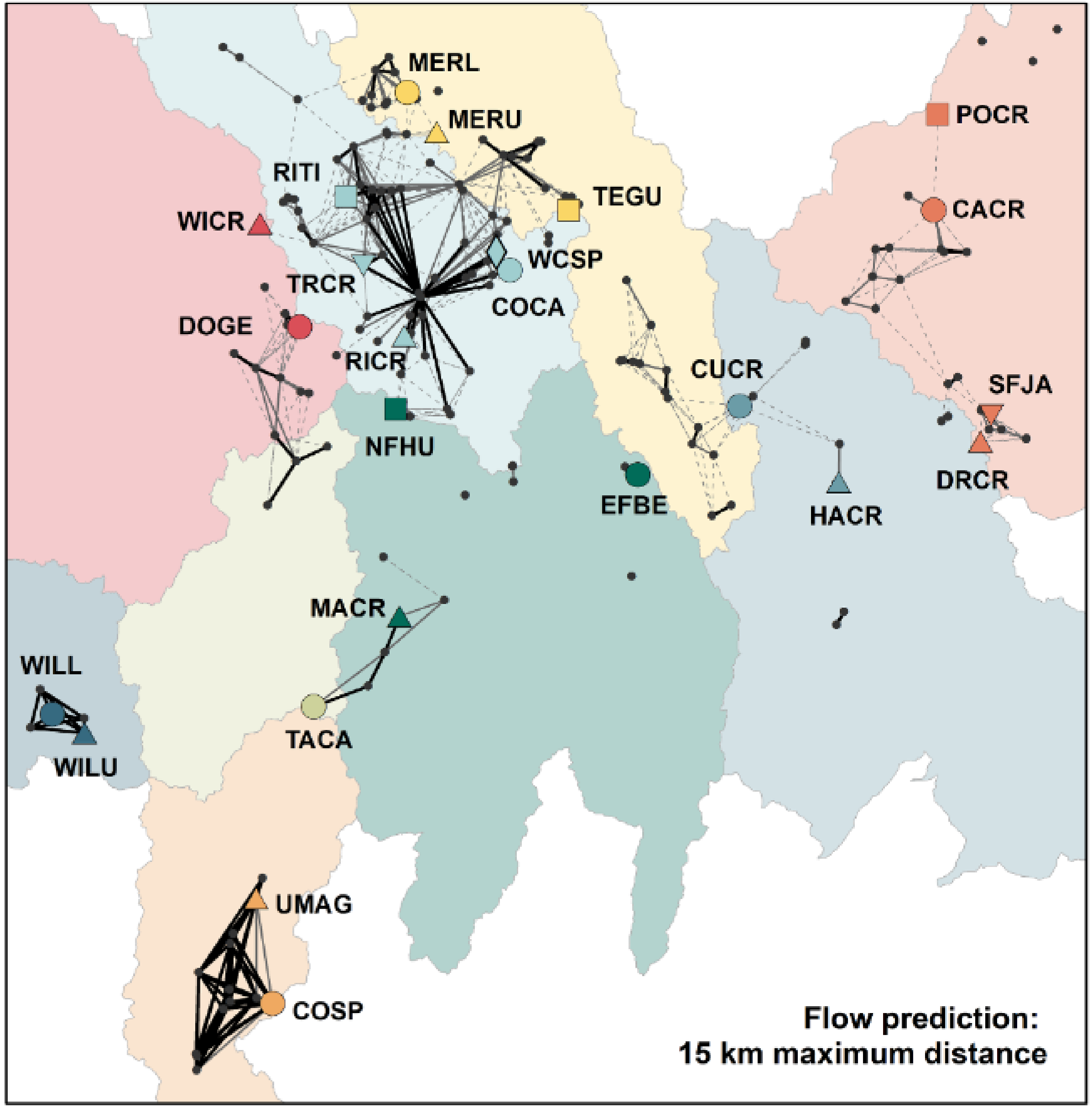


Photo: Joel Sartore

Willow Creek translocation and genetic rescue

- Minimize adaptive differences
- Complementary sourcing based on neutral diversity

Forester et al. 2022, Molecular Ecology

Columbia spotted frog: why is it working?

Case Study:
Successful Collaboration for Columbia Spotted Frog Conservation in Northern and Central Nevada

Kent McAdoo, Natural Resources Specialist, Cooperative Extension
Chad Mellison, Biologist, U.S. Fish and Wildlife Service

>10 year collaboration among multiple federal and state agencies + universities

Conservation agreement and strategy developed and revised and updated frequently

Twice annual meetings with partners to review products and modify analyses in response to new questions/data

Examples: maladaptation, assisted gene flow

2023 marine heatwave reduced extant wild genets from 158 to 23

Elkhorn coral (Acropora palmata) - Threatened

Time series showing an elkhorn coral bleaching and dying in the summer of 2023. The last photo shows the dead skeleton being colonized by filamentous algae. Credit: NOAA Fisheries

Examples: maladaptation, assisted gene flow

PeerJ

Genome-wide survey of single-nucleotide polymorphisms reveals fine-scale population structure and signs of selection in the threatened Caribbean elkhorn coral, *Acropora palmata*

Meghann K. Devlin-Durante* and Iliana B. Baums*

Department of Biology, Pennsylvania State University, University Park, PA, United States of America

* These authors contributed equally to this work.

RESEARCH ARTICLE | APPLIED BIOLOGICAL SCIENCES

Assisted gene flow using cryopreserved sperm in critically endangered coral

PNAS

Mary Hagedorn, Christopher A. Page, Keri L. O'Neil, +13, and Kristen L. Marhaver Authors Info & Affiliations

Edited by Nancy Knowlton, Smithsonian Institution, Washington, DC, and approved August 6, 2021 (received for review June 7, 2021)

September 7, 2021 | 118 (38) e2110559118 | https://doi.org/10.1073/pnas.2110559118

Examples: maladaptation, assisted gene flow

Assisted gene flow from populations in Honduras to promote heat tolerance, disease resistance, and improve recruitment.

There are risks but "…risks of inaction include local extinction and further loss of the coral reef ecosystem functions and services."

CONSERVATION POLICY

Proactive assisted gene flow for Caribbean corals in an era of rapid coral reef decline

Regulatory action could facilitate cross-border efforts to retain ecosystem function

Andrew C. Baker[1], Iliana B. Baums[2,3,4], Sarah W. Davies[5], Andréa G. Grottoli[6], Carly D. Kenkel[7], Sheila A. Kitchen[8], Ilsa B. Kuffner[9], Mikhail V. Matz[10], Margaret W. Miller[11], Erinn M. Muller[12], John E. Parkinson[13], Carlos Prada[14], Andrew A. Shantz[15], Ruben van Hooidonk[16], R. Scott Winters[17,18]

Elkhorn coral: why is it working?

Visible global crisis

Very active scientific community focused on advanced applications of genomic data to address specific threats

Coral conservation in a warming world must harness evolutionary adaptation

To facilitate evolutionary adaptation to climate change, we must protect networks of coral reefs that span a range of environmental conditions — not just apparent 'refugia'.

Madhavi A. Colton, Lisa C. McManus, Daniel E. Schindler, Peter J. Mumby, Stephen R. Palumbi,

Elkhorn coral: why is it working?

CRC WHITE PAPER

Coral Restoration Consortium

Safeguarding Florida's Coral Reefs: The Urgency of Assisted Gene Flow for Elkhorn Coral Conservation

Suggested citation: *CRC Genetics Working Group. (2025). Safeguarding Florida's Coral Reefs: The Urgency of Assisted Gene Flow for Elkhorn Coral Conservation. Coral Restoration Consortium. DOI: 10.5281/zenodo.14920439*

Florida Fish and Wildlife Conservation Commission

Active collaboration among researchers & managers

United States Department of Commerce

National Oceanic and Atmospheric Administration

National Marine Fisheries Service

March 2015

Recovery Plan

Elkhorn Coral *(Acropora palmata)* and Staghorn Coral *(A. cervicornis)*

Southeast Regional Office
Protected Resources Division
263 13th Avenue South
Saint Petersburg, Florida 33701
Phone (727) 824-5312

Staghorn Coral *(Acropora cervicornis)*

Elkhorn coral *(A. palmata)*

Examples: large-effect loci

Tasmanian devil facial tumor disease – declines of 80-90% since first detection in mid 1990s

Photo: Currumbin Wildlife Sanctuary

nature communications

ARTICLE

Received 31 Mar 2016 | Accepted 20 Jul 2016 | Published 30 Aug 2016 DOI: 10.1038/ncomms12684 OPEN

Rapid evolutionary response to a transmissible cancer in Tasmanian devils

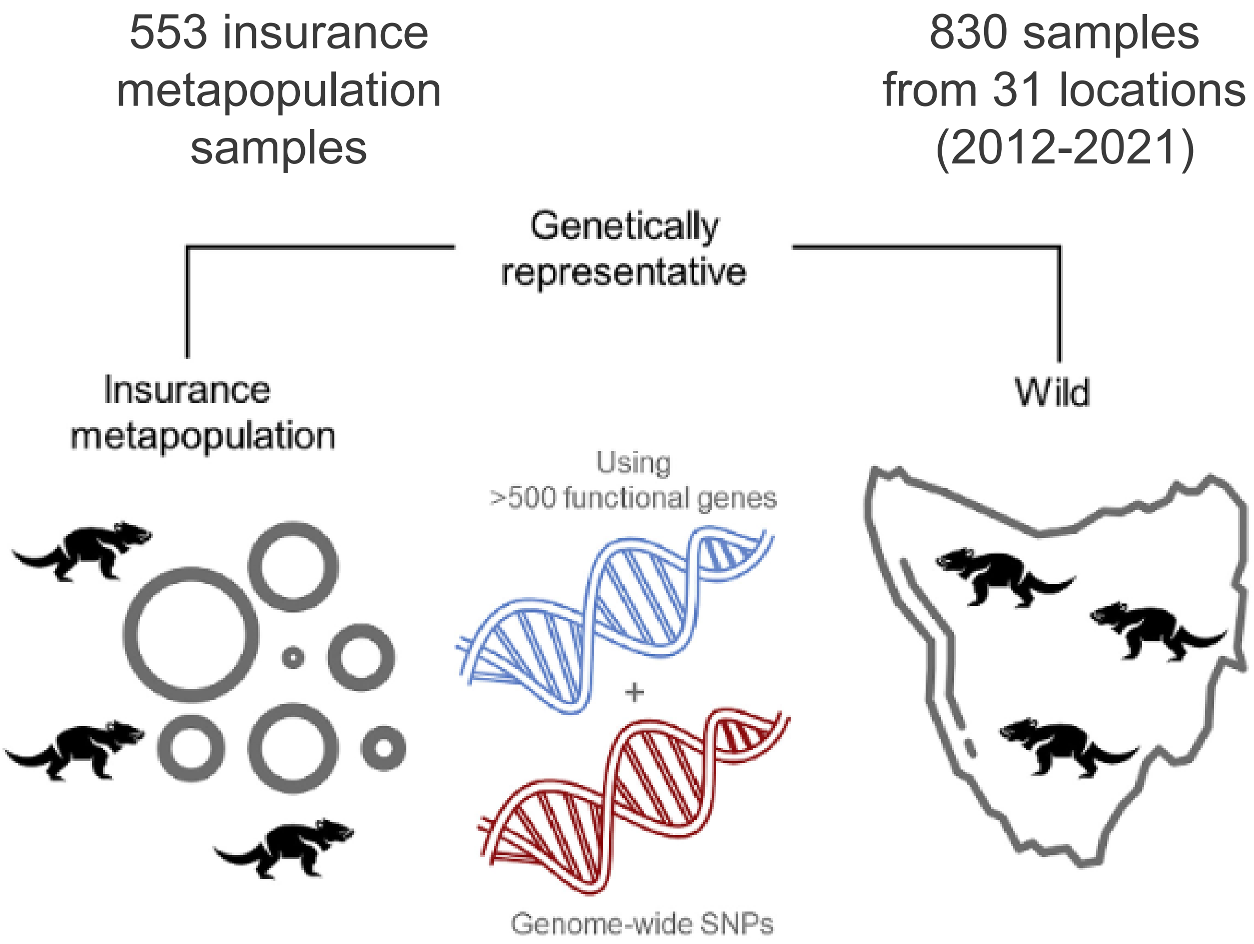
Brendan Epstein[1], Menna Jones[2], Rodrigo Hamede[2], Sarah Hendricks[3], Hamish McCallum[4], Elizabeth P. Murchison[5], Barbara Schönfeld[2], Cody Wiench[3], Paul Hohenlohe[3,*] & Andrew Storfer[1,*]

Two small genomic regions identified in rapid (~4-6 generation) response to disease

Examples: monitor fitness-related alleles

Farquharson et al. 2022, *iScience*

Photo: Currumbin Wildlife Sanctuary



Forms basis for ongoing monitoring and to inform reintroduction strategies

Tasmanian devil: why is it working?

Highly visible/rapid population decline; high public interest and support

Long-term, large scale collaboration across many research labs, zoos, federal agencies, nonprofits, and members of the public

Australian Government

Australian Research Council

SAVE THE TASMANIAN DEVIL PROGRAM

TOLEDO ZOO & AQUARIUM

How do we continue to advance applications …the human side of the work

- Early collaboration with practitioners & maintenance of relationships; willingness to be flexible
- Listen to practitioner needs, then help them determine what is possible with a genomics study (understanding that they may not know what is possible)
- Align sampling design & analyses with practitioner goals – careful design can often mean you can meet their needs while doing additional analyses (be strategic)
- Be transparent about uncertainties and limitations (without undermining the value of the work)

How do we continue to advance applications ...the human side of the work

- Collaborate across research groups; where possible and needed, build consortia dedicated to solving specific conservation problems
- The public can be a great ally to leverage support for conservation action

SAVE THE TASMANIAN DEVIL PROGRAM

How do we continue to advance applications …the policy side of the work

- Take time to understand the policies under which conservation of your species is taking / may take place
- More may be possible than you think – but only if you understand the playing field for practitioners and the breadth of options available for management

Conservation Genetics (2019) 20:115–134
https://doi.org/10.1007/s10592-018-1096-1

REVIEW ARTICLE

Improving conservation policy with genomics: a guide to integrating adaptive potential into U.S. Endangered Species Act decisions for conservation practitioners and geneticists

W. C. Funk[1,2] · Brenna R. Forester[1] · Sarah J. Converse[3] · Catherine Darst[4] · Steve Morey[5]

How do we continue to advance applications …the policy side of the work

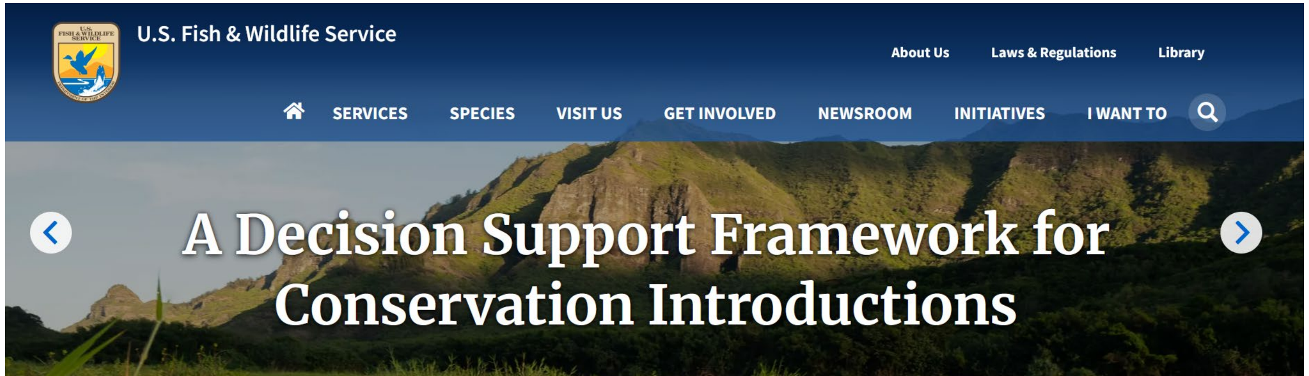
- Many assume conservation legislation is by nature adverse to taking risks, but this is not always the case.
- For example, translocations have been instrumental in the recovery and delisting of 30% of species under the ESA

Conservation Science and Practice
Open Access
A journal of the Society for Conservation Biology

U.S. conservation translocations: Over a century of intended consequences

Ben J. Novak, Ryan Phelan, Michele Weber

First published: 05 March 2021 | https://doi.org/10.1111/csp2.394 | Citations: 46



How do we continue to advance applications …the science-policy nexus

- In many contexts, it is not feasible to wait for perfect (or even better) information, so costs and benefits must be carefully considered

FRONTIERS IN ECOLOGY and the ENVIRONMENT

Review

Counting the books while the library burns: why conservation monitoring programs need a plan for action

David B Lindenmayer, Maxine P Piggott, Brendan A Wintle

First published: 11 November 2013 | https://doi.org/10.1890/120220 | Citations: 185

How do we continue to advance applications …the science-policy nexus

"…where fast decisions are needed, for example in the case of highly threatened species with rapidly declining populations, genomic offset might be one of the only possibilities to inform conservation management. In such scenarios, the trade-off between the ongoing development of the approach and the urgent need for action has to be carefully discussed." – Rellstab et al. 2021, *Evolutionary Applications*

Guam kingfisher / sihek (*Todiramphus cinnamominus*) – Endangered. The single extant population in the wild is due to a conservation introduction outside of its native range. Photo: FWS

Thank you!

Questions?

Joshua tree (*Yucca* sp.) - Under assessment; photo: Bernard Gagnon